

Enhancing Microrobot Swarm Stability and Adaptation by Autonomous Field-of-View Planning

Zhaowen Su, Lijun Fang, Hoyeon Kim, and U. Kei Cheang*

Automatic navigation of microrobot swarms driven by magnetic fields has attracted considerable attention due to potential applications in biomedical fields. However, achieving minimal loss and maintaining swarm cohesion while traversing heterogeneous landscapes over long distances remains a challenge. This article introduces a control strategy based on autonomous field-of-view (FOV) planning for navigating microrobot swarms across large workspaces that span multiple FOVs. High-resolution global images of the workspace are obtained using an image stitching method that combines phase correlation and template matching. Global path planning is accomplished with the A* algorithm, followed by local path planning utilizes the optimized informed rapidly-exploring random tree star (OI-RRT*) algorithm in each FOV to ensure swarm adaptation. The strategy also incorporates an FOV planning algorithm to optimize FOV positioning, along with a displacement platform to ensure smooth transitions between FOVs. A real-time visual feedback control system monitors both channel width and swarm position. This strategy improves swarm navigation efficiency and stability, as demonstrated through experimental validation, and holds significant potential for targeted drug delivery and other biomedical applications.

1. Introduction

Microrobots hold immense biomedical potential by accessing hard-to-reach areas in the human body, including targeted drug delivery,^[1,2] stem cell transportation,^[3] revascularization,^[4] ocular surgery,^[5] and pollutant removal.^[6] To achieve various functionalities in specific fluidic environments, researchers have explored microrobots with different structures and properties, including helical microswimmers,^[7] achiral microswimmers,^[8] soft

microrobots,^[9] spore-based microrobots,^[10] and even bacteria-hybrid microrobots.^[11] However, controlling single microrobots or a group of uncoordinated microrobots is inefficient for practical applications. To increase efficiency, swarms are emerging as a promising solution due to their ability to reach targets with coordination^[12] and enhance imaging resolution.^[13] Unlike macroscale robots, microrobot swarms currently cannot be equipped with sensors and controllers proportionate to their size; instead, they have to rely on external fields for actuation, such as optical,^[14] acoustic,^[15] electric^[16] and magnetic fields.^[17,18] In comparison, magnetic fields can penetrate deep biological tissues and exhibit better biocompatibility,^[19] which leads to their wide application in the generation, actuation, and control of microrobot swarms.^[20,21]

Precise control of the input magnetic field is essential for guiding microrobot swarms in tasks like targeted drug delivery.

Various methods have been developed to enhance swarm motion control, including statistical-based tracking of vortex swarms,^[22] fuzzy control for elliptical swarm adjustments,^[23] enhanced bidirectional rapidly-exploring random tree star with genetic algorithm-based linear quadratic regulator for dynamic target tracking,^[24] hierarchical radar for obstacle navigation,^[25] and deep learning for adaptation in complex environments.^[26] While these methods enable efficient swarm control for microscale tasks, their application in large workspaces spanning multiple fields-of-view (FOVs) remains unexplored. In scenarios involving large workspaces, microrobot swarms must be capable of navigating autonomously beyond a single FOV.

Using lower magnification increases the FOV but often reduces imaging resolution, which may compromise the accuracy of path planning and autonomous motion control. Therefore, micromazes used in experiments are typically designed to fit within the dimensions of the FOV.^[27] Traditional FOV tracking for microrobots requires frequent adjustments to the camera^[28] and the coil^[29] to align with the microrobot's centroid. However, this method is unsuitable for microrobot swarms, as it requires stationary Helmholtz coils for effective, torque-free swarm control. Moreover, swarm loss during translational motion poses a significant problem that must be considered.^[20] Frequent and rapid FOV tracking movements can disrupt swarm motion and cohesion. Another approach is to manually move the microscope stage to keep

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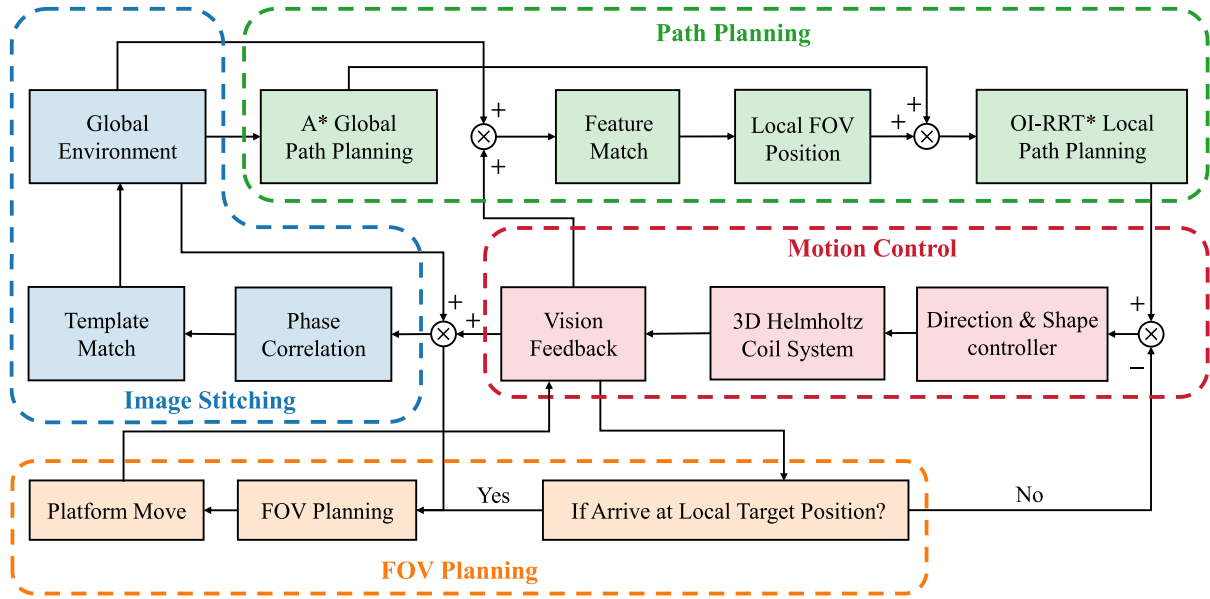


Figure 1. A system diagram that consists of the image stitching unit, path planning unit, motion control unit, and the FOV planning unit.

the microrobot swarm within the FOV.^[26] However, the absence of autonomous FOV optimization for swarm control resulted in suboptimal swarm cohesion and adaptability during long-distance travel in complex environments.

Herein, this article proposes a control scheme that integrates swarm adaptation with autonomous FOV control to effectively navigate environments with varying dimensions across multiple FOVs, addressing the challenge of traversing heterogeneous landscapes with minimal loss and ensuring swarm cohesion over long distances. The overall control scheme, illustrated in **Figure 1**, comprises four main operational components, each responsible for a specific function: path planning, image stitching, motion control, and FOV planning. To acquire high-resolution global images of the workspace, phase correlation and template matching for image stitching are employed. Global path planning is achieved using the A* algorithm, while local path planning is carried out with the optimized informed rapidly-exploring random tree star (OI-RRT*) algorithm. The FOV planning algorithm is implemented to determine the optimal position for the next FOV. A displacement platform moves between FOVs, and feature point matching is used to localize the local FOV within the global map. The control of the microrobot swarm is supported by a visual feedback control system, which continuously monitors the channel width and the swarm's position in real time.

2. Control Scheme

2.1. Image Stitching

To efficiently construct a global map, a hybrid image stitching method integrating phase correlation and template matching was employed, as illustrated in **Figure 2**. This approach identifies, aligns, and merges overlapping regions between image pairs, enabling the generation of a panoramic view. The image

was first converted to grayscale. The phase correlation method^[30] was then applied to estimate the relative translation between overlapping regions, accounting for planar movement between the camera and the working surface.

Assuming $f_1(x,y)$ and $f_2(x,y)$ are two image signals captured by cameras at different positions, which are shown in Figure 2a, and $f_2(x,y)$ was obtained by translating $f_1(x,y)$ by (dx,dy) , this relative displacement can be expressed as

$$f_2(x,y) = f_1(x-dx, y-dy) \quad (1)$$

According to the properties of the Fourier transform, a shift in the image domain is equivalent to a phase change in the Fourier domain. The spectra of the two images have the following relationship

$$F_{2(u,v)} = F_{1(u,v)} e^{-j2\pi(udx+vd y)} \quad (2)$$

where u and v denote the positions of the images in Fourier space. $F_{1(u,v)}$ and $F_{2(u,v)}$ are the Fourier transforms of $F_{1(x,y)}$ and $F_{2(x,y)}$ respectively. The normalized cross-power spectrum $H(u,v)$ corresponding to the two images is defined as:

$$H(u,v) = \frac{F_1(u,v)F_2^*(u,v)}{|F_1(u,v)F_2^*(u,v)|} = e^{-j2\pi(udx+vd y)} \quad (3)$$

where the superscript $*$ denotes complex conjugation. Thus, the phase of the normalized cross-power spectrum corresponds to the phase difference between the two images. Computing the Fourier inverse transform of the normalized cross-power spectrum yields the phase correlation image, which is a pulse-like image $C(x,y)$. In this image, most pixel values are close to zero, except at positions representing displacement. Therefore, the relative displacement between the two images can be determined by identifying the position of the maximum value $C_{max}(x,y)$, as illustrated in Figure 2b.

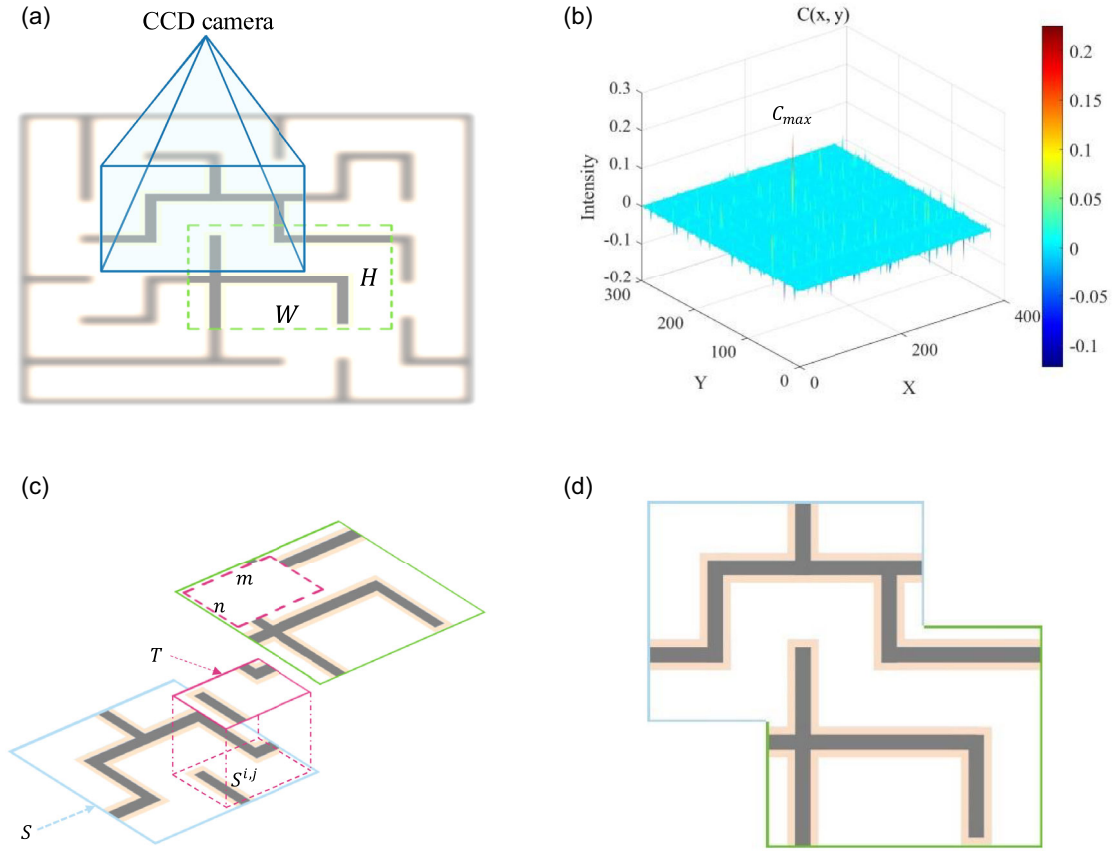


Figure 2. The schematics of image stitching. a) Local high-resolution image acquisition, where W and H represent the width and height of the image, respectively. b) The impulse response $C(x, y)$ of the image to be stitched using phase correlation. c) The schematics of template matching. T is the selected template, with width and height denoted as m and n , respectively. S is the search region, with the subimage S^{ij} being the search image under the template. d) Stitching result.

Phase correlation is sensitive to lighting variations and device-related photoelectric noise, which may lead to inaccuracies in the calculated relative displacement. As a result, it is primarily used to approximate the overlapping regions of images. To achieve precise image registration and stitching, template matching^[31] is employed. The position and dimensions of the template $T(C_x, C_y, m, n)$ are determined based on $C_{max}(x_{max}, y_{max})$. Here, c_x and c_y represent the coordinates of the template's top-left corner, while m and n denote its width and height, respectively.

$$T(c_x, c_y, m, n) = \begin{cases} T(-x_{max}, -y_{max}, W + x_{max}, H + y_{max}), & y_{max} < 0, x_{max} < 0 \\ T(-x_{max}, 0, W + x_{max}, H - y_{max}), & y_{max} \geq 0, x_{max} < 0 \\ T(0, -y_{max}, W - x_{max}, H + y_{max}), & y_{max} < 0, x_{max} \geq 0 \\ T(0, 0, W - x_{max}, H - y_{max}), & y_{max} \geq 0, x_{max} \geq 0 \end{cases} \quad (4)$$

Here, W and H respectively denote the width and height of the images to be stitched. After obtaining the template T , it is overlaid and translated across the search region S .

The segment of S covered by T is termed the subimage S^{ij} , where (i, j) represents the coordinates of the top-left corner of S^{ij} within the search region S , as shown in Figure 2c. Given that S has dimensions $W \times H$ and T has dimensions $m \times n$, the indices i and j are bounded by $0 < i \leq W - m$ and $0 < j \leq H - n$, respectively.

Images captured by the same color camera from different positions contain overlapping regions. Thus, it is expected that a subimage S^{ij} perfectly matching template T . Ideally, the difference between T and S^{ij} should be zero. However, due to factors such as lighting variations and device-related photoelectric noise, a perfect match is unattainable. To quantify the similarity between T and S^{ij} , the normalized sum of squared difference (SSD) is used. By moving the template to $(W - m) \times (H - n)$ positions and computing the maximum SSD, the location (i, j) , where the maximum is achieved, represents the optimal matching point.

2.2. Path Planning Algorithm

After obtaining the complete image of the microchannel, the A* algorithm is utilized to compute the global path. To account for the physical dimensions of the swarm, a configuration space

is added around the walls to prevent collisions between the swarm and the boundaries. In the A* algorithm framework, the global image is represented as a grid of nodes, with each node assigned a priority based on the following function

$$f(n) = g(n) + h(n) \quad (5)$$

where $g(n)$ represents the cost of node n from the starting point and $h(n)$ denotes the estimated cost of node n to the destination, serving as the heuristic function in the A* algorithm. The Manhattan distance is employed as the heuristic function. The overall priority of node n is defined by $f(n)$, which combines $g(n)$ and $h(n)$. During traversal, the algorithm selects the node with the highest priority (i.e., the smallest $f(n)$) until the destination is reached, thereby establishing an optimal global path. This global path planning enables the microrobot swarm to adapt its morphology for channels with varying widths (Figure 3) and navigate complex pathways (Figure 4).

The path planning strategy coordinates swarm motion modes through three-stage channel analysis: 1) During global path generation, the A* algorithm identifies channel width variations along the trajectory using prescanned dimensional data; 2) deformation-critical waypoints ($1 \times$ swarm radius before narrow channels) are embedded into the path through cost-map adjustments in the OI-RRT* local planner; and 3) real-time visual feedback synchronizes swarm configuration transitions with these predefined waypoints. This hierarchical planning approach categorizes paths into two motion regimes, red segments (swarm-width-compatible channels) enable direct translation, while green segments (narrower channels) trigger deformation at specified locations (stored as metadata in path coordinates). The critical coordination manifests in temporal-spatial alignment between swarm deformation and path progression: 1) Narrowing transitions (red $\rightarrow$ green) initiate ellipsoidal reshaping $1 \times$ swarm radius upstream through predictive path parameter interpolation and 2) widening transitions (green $\rightarrow$ red) delay reconfiguration until complete narrow-channel traversal via path-length-dependent triggering, as shown in Figure 3.

To enable microrobot swarms to navigate over long distances, it is necessary to efficiently generate the desired local paths within each FOV after acquiring the global path. Therefore, we propose an OI-RRT* algorithm, which outperforms RRT* in terms of search speed due to its restriction of the

search area within an elliptical region, as illustrated in Figure 4b–d. The OI-RRT* algorithm also allows for the adaptation to unforeseen obstacles (Figure S1 and Video S5). First, as described in Algorithm 1, feature point matching is used to align the global image with the swarm's FOV. We employ the scale-invariant feature transform^[32] to extract key points. Then, in the overlapping regions, matching key points are identified by comparing their Hamming distances using the fast library for approximate nearest neighbors.^[33] This alignment enables the mapping of the global path onto the local FOV.

The endpoint of the mapped local path is extracted as the target for local path planning, while the point closest to the swarm serves as the starting point. Beginning at the root node (χ_{start}), a random node (χ_{rand}) is sampled within the local FOV. To improve path exploration efficiency, there is a probabilistic bias toward sampling along the line connecting the start and endpoint. Next, all nodes in the tree are traversed to find the node (χ_{near}) with the minimum cost relative to χ_{rand} . Using the Steer function, an extension point (χ_{new}) is generated in the direction of χ_{rand} , constrained by a specified step length. Collision detection is then performed on χ_{new} . If no collisions are detected, χ_{near} is assigned as the parent of χ_{new} , establishing a connection between the two nodes.

The next step involves using the current node χ_{new} as the center to generate a neighboring region with a given radius. All points within this region will be reconnected by Rewire functions in Algorithm 1 to ensure that these points have the minimum distance to the start point in the tree. Once the tree nodes reach the endpoint, an initial path is established. Utilizing the length of this initial path as the major axis c_{best} and the straight-line distance between the start and end points as the focal length c_{min} , an elliptical range can be delineated^[34] (i.e., cyan ellipse in Figure 4b)

$$\tan \theta = \frac{\chi_{end}^y - \chi_{start}^y}{\chi_{end}^x - \chi_{start}^x} \quad (6)$$

$$\frac{((x - x_c) \cos \theta + (y - y_c) \sin \theta)^2}{c_{best}^2} + \frac{((x - x_c) \sin \theta + (y - y_c) \cos \theta)^2}{c_{best}^2 - c_{min}^2} = 1 \quad (7)$$

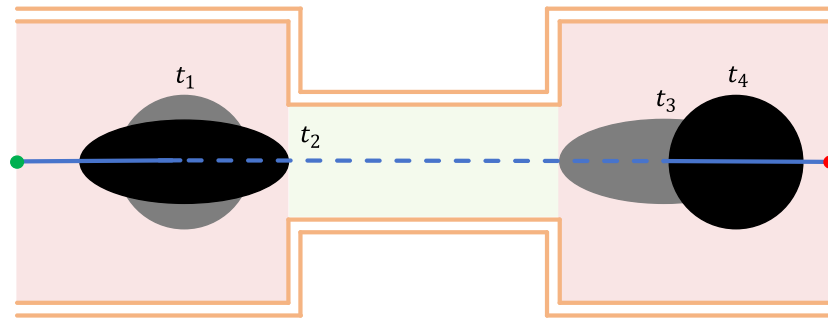


Figure 3. Schematic of the swarm deforming to traverse a narrow channel. The dashed line represents the elliptical swarm path, the solid line represents the vortex swarm path, and the intersection points between the dashed and solid lines are the swarm deformation points.

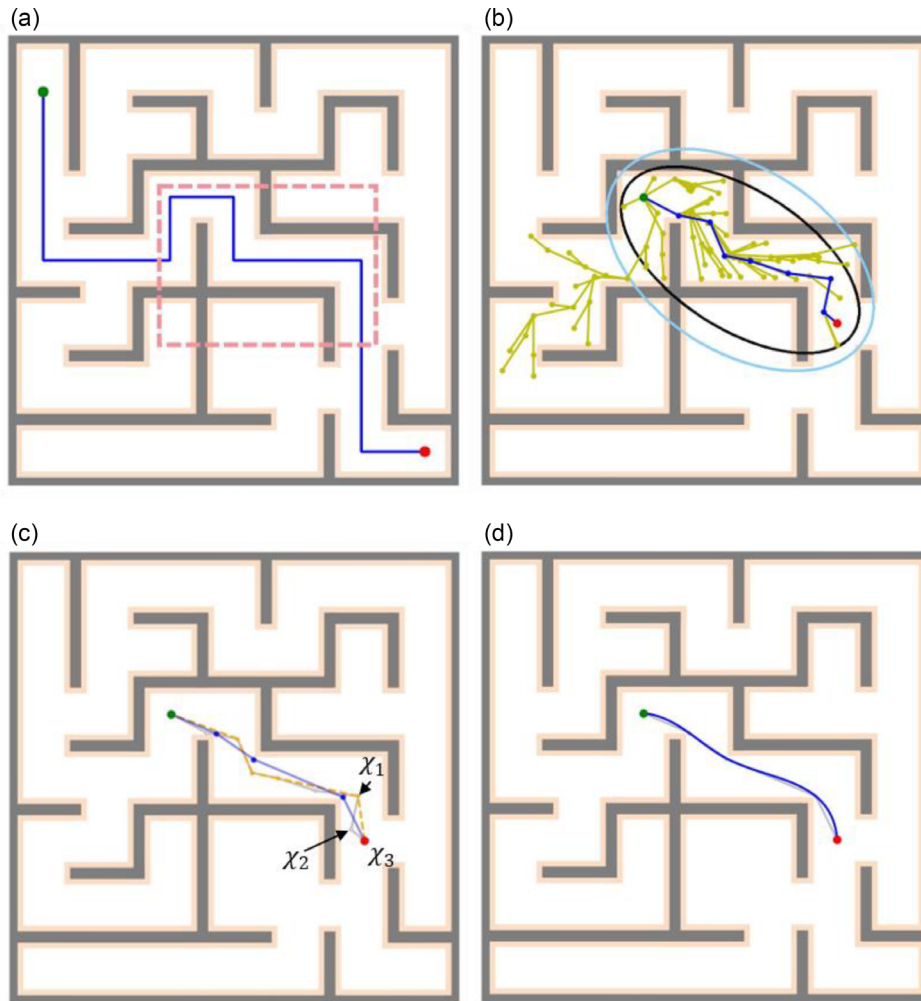


Figure 4. The schematics of the global A* and local OI-RRT* path planning algorithm. a) Global path generation, where the red box represents the local FOV. b) Generation of local initial path. Yellow lines are branches of the tree and blue line is effective path. New tree branches are sampled within a continuously shrinking ellipse once the initial path is generated. c) Path optimization by the TriangleInequality (orange lines) and ObstacleSampling (blue lines) functions in Algorithm 1. x_1, x_2, x_3 form a triangle. d) Path interpolation by Interpolation function in Algorithm 1.

where x_c and y_c represent the coordinates of the center of the ellipse, namely the coordinates of the midpoint of the line connecting the start point x_{start} and the endpoint x_{end} . Then sampling is confined within the elliptical range, with the range continuously narrowing (i.e., from cyan ellipse to black ellipse in Figure 4(b)) as the path undergoes iterative optimization. The path is optimized through the triangle inequality^[35] by the TriangleInequality function in Algorithm 1, where the third side is smaller than the sum of two sides (i.e., $|\overrightarrow{x_1x_3}| < |\overrightarrow{x_1x_2}| + |\overrightarrow{x_2x_3}|$), then $|\overrightarrow{x_1x_3}|$ forms a new path, thereby reducing the path length, as illustrated by the orange dashed line in Figure 4(c). To further optimize, sampling is performed near obstacles along the path by the ObstacleSampling function in Algorithm 1, as indicated by the blue line in Figure 4(c), reducing the path length. Finally, in order to enhance the swarm's trajectory tracking capability, B-spline curves are used for path smoothing interpolation by the Interpolation function in Algorithm 1, as demonstrated in Figure 4(d).

2.3. Platform Moving Strategy

To enable the effective operation of microrobot swarms in large workspaces, a mobilized platform was introduced to transport the controlled entities. However, the movement of this platform can cause disturbances to the microrobot swarm, potentially leading to its disassembly under severe conditions. Thus, minimizing the platform's movement frequency is critical. To address this challenge, a FOV planning algorithm was developed and implemented, as illustrated in Figure 5. The algorithm begins by establishing a safety frame to ensure the swarm remains within the FOV. This safety frame defines the optimal range within the FOV where the swarm should operate. Only paths that fall entirely within the safety frame (depicted as green frames in Figure 5(a)) are considered valid. The safety frame is mathematically defined as

$$f(h, w) = \mathcal{F}(H - \epsilon l_s, W - \epsilon l_s) \quad (8)$$

Algorithm 1: OI-RRT* Algorithm

```

 $F_p \leftarrow \text{FLANN}(\text{sift}(l_{\text{local}}), \text{sift}(l_{\text{global}}));$ 
 $A_{\text{local}} \leftarrow \text{Find\_LocalArea}(\text{best}(F_p), );$ 
 $P_{\text{local}} \leftarrow \text{Find\_LocalPoints}(A_{\text{local}}, P_{\text{global}});$ 
 $X_{\text{start}} \leftarrow \text{GetNearest}(P_{\text{local}}, X_{\text{swarm}});$ 
 $X_{\text{end}} \leftarrow \text{GetLast}(P_{\text{local}});$ 
 $\text{CollisionBufferLayer}();$ 
 $T = \text{InitTree}(X_{\text{start}});$ 
for  $n = 1$  to  $N$  do
     $C_{\text{best}} \leftarrow \min(\text{cost}(\text{PathFound}()));$ 
     $x_{\text{rand}} \leftarrow \text{Sample}(X_{\text{start}}, X_{\text{end}}, C_{\text{best}});$ 
     $x_{\text{new}} \leftarrow \text{Steer}(x_{\text{rand}}, x_{\text{near}});$ 
    if  $\text{CollisionFree}(x_{\text{new}}, x_{\text{near}})$  then
         $x_{\text{neighbor}} \leftarrow \text{Neighbor}(x_{\text{new}}, T, r);$ 
         $T \leftarrow \text{Rewire}(x_{\text{neighbor}}, x_{\text{new}}, T);$ 
    if  $\text{PathFound}()$  then
         $T_t \leftarrow \text{TriangleInequality}(X_{\text{start}}, X_{\text{end}}, T);$ 
         $T_o \leftarrow \text{ObstacleSampling}(X_{\text{start}}, X_{\text{end}}, T_i);$ 
         $T_i \leftarrow \text{interpolation}(X_{\text{start}}, X_{\text{end}}, T_o);$ 
Return  $T_i$ 

```

where $\mathcal{F}$ is the FOV, H and W are the height and width of the FOV respectively, h and w are the height and width of the safety frame respectively, l_s is the radius of the detected swarm, and ε is a constant between 2 and 2.5 used to compensate for detection errors. The number of movements ξ can be determined by the following equation

$$\mathcal{L}_{\text{global}} = \sum_{k=1}^{\xi} \mathcal{L}_{\text{local}}^k, \mathcal{L}_{\text{local}}^i \in C \quad (9)$$

Here, $\mathcal{L}_{\text{global}}$ represents the total distance of the global path, and $\mathcal{L}_{\text{local}}^i$ denotes the local length of the global path that can be accommodated within the local FOV in each instance, C denotes the set of continuous functions. In image processing, continuous paths are composed of discrete points

$$\mathcal{L} = \sum_{i=1}^N \sqrt{(\sigma_x^{i+1} - \sigma_x^i)^2 + (\sigma_y^{i+1} - \sigma_y^i)^2} \quad (10)$$

where σ^i denotes the coordinates of the i path point, N represents the length of the path. To minimize the number of movements ξ , each segment of local path points should be mutually exclusive

$$\sigma^i \cap \sigma^j = \emptyset, \forall \sigma^i \in \mathcal{L}_{\text{local}}^k, \sigma^j \in \mathcal{L}_{\text{local}}^t \quad (11)$$

where $k \neq t$, $k, t \in [1, \xi]$. To achieve mutually exclusive, the endpoint of the previous segment of the path can be used as the starting point for the next segment of the path, as illustrated by the function `GetLastPoint` in **Algorithm 2**. Then it is necessary to maximize the length of continuous local paths of the

global path $\mathcal{L}_{\text{local}}^k$ included in each movement, and the local path points satisfy the following conditions within the local FOV

$$\begin{cases} w - \max\{\sigma_x^i - \sigma_x^j\} \geq 0, \\ h - \max\{\sigma_y^i - \sigma_y^j\} \geq 0, \end{cases} i \neq j \quad (12)$$

where $i, j \in [1, N]$. The condition is checked using the function `PathExceedFOV` in **Algorithm 2** to achieve path segmentation (shown as red waypoints in Figure 5(a)). After segmenting each local path, it would indicate that multiple FOVs contain that segment of the local path, as shown by the gray dashed box in Figure 5(b) if Equation (12) is not equal to zero. These FOVs are found using the function `FindFOVsIncludePath` in **Algorithm 2**.

In order to minimize the impact on the microrobot swarm, the distance of each displacement platform movement should be kept as small as possible. That is, the distance D between the center of the next FOV and the current FOV should be minimized

$$D = \min\left\{\sqrt{(f_x^{k+1} - f_x^k)^2 + (f_y^{k+1} - f_y^k)^2}\right\} \quad (13)$$

where $\mathcal{L}_{\text{local}}^k \subset f^k$, $\mathcal{L}_{\text{local}}^{k+1} \subset f^{k+1}$, $k \in (0, \xi)$. The nearest FOV is found using the `Nearest` function in **Algorithm 2**. Without the FOV Planning Algorithm, the FOV would simply move to the center of the swarm, requiring 9 platform movements to follow the swarm through the maze, resulting in a total movement of 1364 μm , as shown in Figure 5(c). In contrast, with the FOV Planning Algorithm, only 4 platform movements are required to reach the endpoint, totaling 1008 μm . This represents a 26% reduction in movement distance and more than a 50% reduction in the number of movements, as illustrated in Figure 5(d), significantly reducing both the time and distance of displacement platform movements.

2.4. Vision Feedback Control

To achieve precise local path following, vision-based feedback motion control is essential. Real-time acquisition of both local and global positions of the microrobot swarm is required for effective motion control. Target recognition of the swarm within the FOV is critical for obtaining motion information. During target recognition, threshold segmentation is applied to the microscope images. However, the microscopic environment contains numerous noise points with volumes similar to that of the swarm, which interfere with accurate recognition. To address this issue, a region of interest (ROI) tracking method is adopted, restricting target recognition to a smaller, focused area, thereby improving accuracy and efficiency, while a deformation-aware mechanism further enhances robustness by adaptively adjusting the window size based on a minimum bounding ellipse fitted to the swarm's shape.

In the ROI tracking target detection, the center coordinates $p_l(0, 0)$ of the first frame of swarm movement are determined. These coordinates are in the pixel coordinate system of the current FOV. A rectangular window is then drawn based on the center coordinates of the first frame of swarm movement. Since ROI tracking is a posterior window tracking strategy,

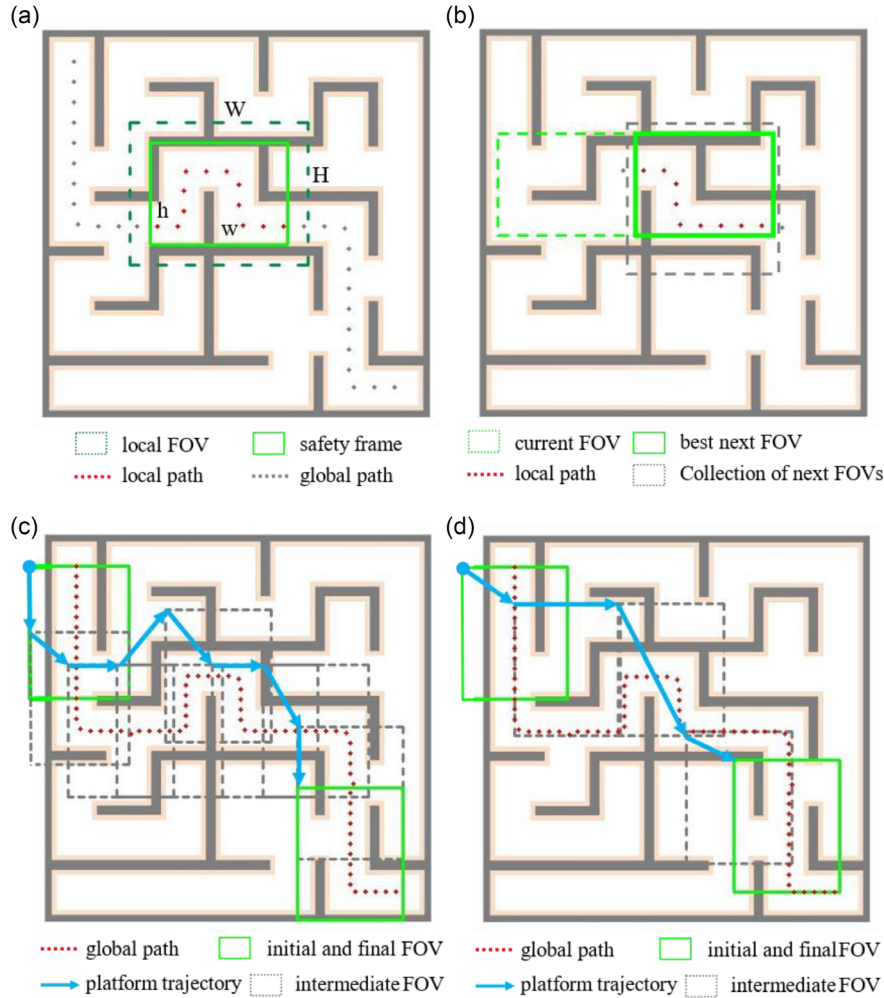


Figure 5. The FOV planning strategy. a) Local longest path segmentation. Local FOV with height and width denoted as W and H . Safety frame with height and width referred to as h and w . b) Nearest FOV Selection. The collection of next FOVs containing the local path $\mathcal{L}_{local}^i$, while the current FOV represents f^k , and the best next FOV represents f^{k+1} . c) Platform displacement planning by moving the swarm to the center of the FOV. d) Platform displacement planning by FOV Planning Algorithm.

Algorithm 2: FOV Planning Algorithm

```

 $P_{local} \leftarrow FindLocalPath(F_{current}, P_{global});$ 
 $I_{last} \leftarrow GetLastPoint(P_{local});$ 
for  $n = I_{last}$  to  $length(P_{global})$  do
    if  $PathExceedFOV(I_{last}, I_n)$  then
         $P_{next} \leftarrow Path((I_{last}, I_{n-1}));$ 
 $\mathcal{F} = FindFOVsIncludePath(P_{next});$ 
 $F_{next} = Nearest(\mathcal{F}, F_{current});$ 
Return  $F_{next}$ 

```

the window tracking lags behind the swarm movement. It is necessary to ensure that the swarm does not move out of the rectangular window within one frame. Thus, the size of the rectangular window is determined by the preset movement speed of

the swarm. Then, the ROI is binarized, and the swarm center is calculated using the first moment.

By calculating the distance d_i from the swarm center to the ROI center and vectorially adding it to the known current ROI center coordinates $p_l(k, i)$, the center position $p_l(k, i + 1)$ of the ROI for the next frame can be obtained, as shown in **Figure 6**. A rectangular window is then drawn at this position, continuously keeping the swarm centered in the ROI. By using the recursive method, the local coordinates of the swarm within the current FOV can be obtained

$$p_l(k, i) = \sum_{j=1}^{j=0} d_j + p_l(k, 0) \quad (14)$$

The coordinates of the cluster on the global map can be obtained by vectorially adding the coordinates of the FOV and the local coordinates

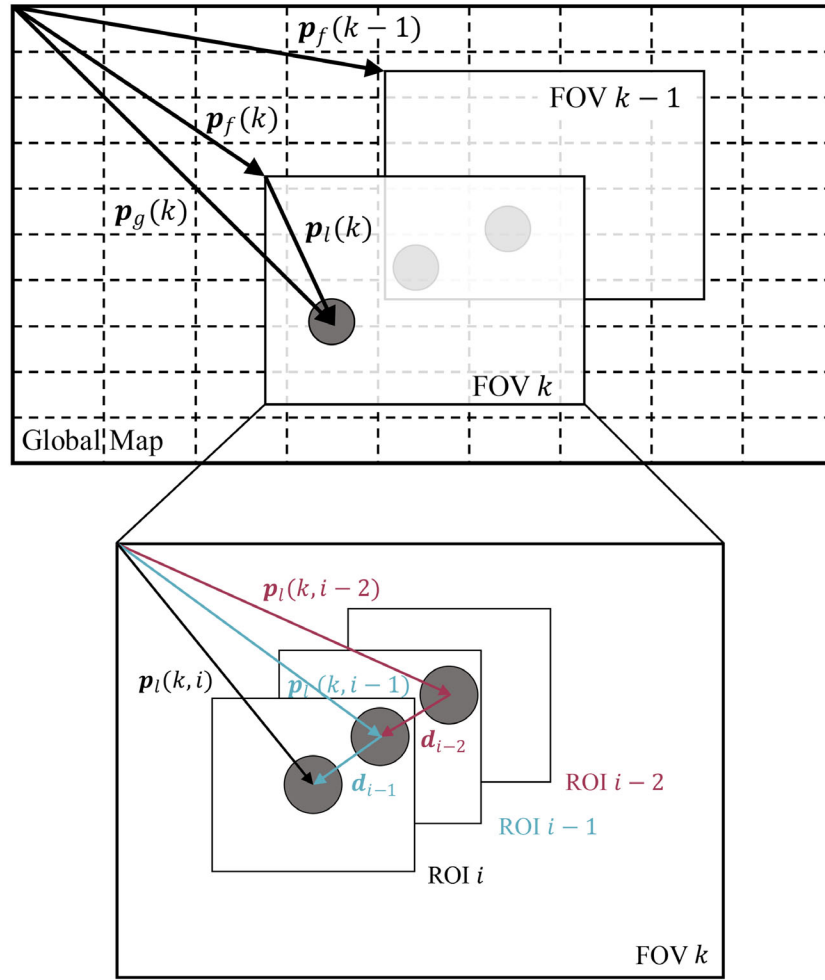


Figure 6. The relationship between the ROI coordinates, FOV coordinates, and the local and global coordinates of the swarm.

$$p_g(k, i) = p_f(k) + p_l(k, i) \quad (15)$$

2.5. Motion Control

To actuate the omnidirectional motion of the nanoparticle chains as a compact swarm, an elliptical magnetic field is applied, as shown in the inset of **Figure 7**. The field can be expressed as follows

$$B_{Omni} = R(\gamma) \cdot R(\beta) \cdot B_E(t) \quad (16)$$

In (19), $B_E(t)$ is the elliptical magnetic field, $R(\beta)$ provides a pitch angle β for the plane of the magnetic field, $R(\gamma)$ is for determining the long-axial direction of the swarm, and B_E along with $R(\gamma)$ collectively determine the direction of the motion of the swarm. The above functions are expanded as:

$$B_E(t) = [a \cos(\omega t) \quad b \sin(\omega t) \quad \sin(\omega t + \alpha)] \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \\ \hat{e}_z \end{bmatrix} \quad (17)$$

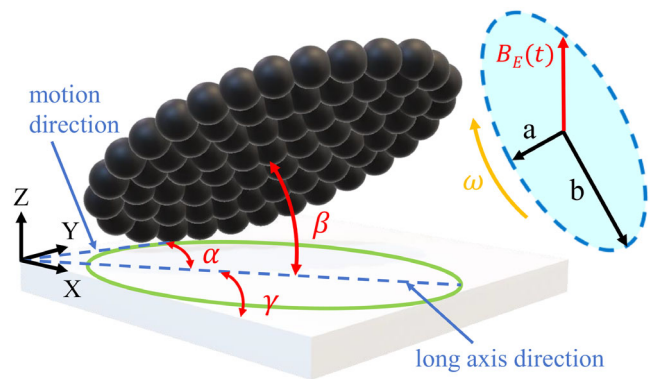


Figure 7. Motion control of swarm using an elliptical magnetic field.

$$R(\beta) = \begin{bmatrix} \cos(\beta) & 0 & 0 \\ 0 & \cos(\beta) & 0 \\ 0 & 0 & \sin(\beta) \end{bmatrix} \quad (18)$$

$$R(\gamma) = \begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (19)$$

where ω is the angular velocity of the elliptical magnetic field, t denotes time, a and b are the minor and major axes of the elliptical field, respectively. The unit vectors $\hat{e}_x$, $\hat{e}_y$, and $\hat{e}_z$ are mutually perpendicular. The yaw angle γ determines the long-axis orientation of the swarm, while $\gamma + \alpha$ specifies the swarm's motion direction, specifically tangent to the substrate. By adjusting the pitch angle β , the swarm's motion speed can be controlled. The long axis of the elliptical magnetic field is oriented perpendicular to the long axis of the swarm, as illustrated in Figure 7. This alignment is due to the stronger magnetic field intensity along the long axis of the elliptical field, which promotes a more compact aggregation of nanoparticles in that direction, compared to the weaker intensity along the minor axis. Additionally, the rotation direction of the elliptical magnetic field is determined by the initial phase settings of $\hat{e}_x$ and $\hat{e}_y$. This rotation direction influences the driving force acting on the swarm, thereby affecting its motion direction.

The omnidirectional motion of the microrobot swarm can be demonstrated by controlling the swarm to move along the set trajectory using visual position feedback. The speed of the swarm's movement is controlled by the pitch angle β . However, excessive pitch angles can render the swarm unstable, hence maintaining a constant pitch angle β during movement is necessary. Moreover, to ensure that the swarm moves along the long axis, α is set to zero, and the swarm's yaw angle γ is adjusted to follow the set trajectory. During motion control, the angle between the set point and the center position of target recognition is calculated, and this angle is set as the yaw angle of the magnetic field:

$$\gamma = \arctan\left(\frac{y_{tar} - y_{cur}}{x_{tar} - x_{cur}}\right) \quad (20)$$

where x_{tar} and y_{tar} are the pixel coordinates of the set target, x_{cur} and y_{cur} denote the pixel coordinates obtained after the swarm undergoes target recognition. Through motion control based on visual position feedback, the magnetic field in three directions is calculated in real-time according to (17), enabling the swarm to move along the set trajectory.

3. Experimental Section

3.1. System Setup

The magnetic actuation system is depicted in Figure 8. We employed a three-axis Helmholtz coil system, which was operated via LabVIEW and Python to generate the elliptical magnetic field for the actuation of microrobot swarms. Three power supplies (Kepco BOP20-5M) supply power to the system, while a National Instruments data acquisition device, with its PCI card (PCI-6259), manages control signals. Vision feedback was facilitated by a CMOS camera (FLIR BFS-U3-13Y3M-C) with a FOV of $3072 \times 2458 \mu\text{m}$, along with a host computer to implement the control algorithm. Additionally, a motor-controlled mobile platform (Thorlabs MP100 and PLS-XY) served dual purposes:

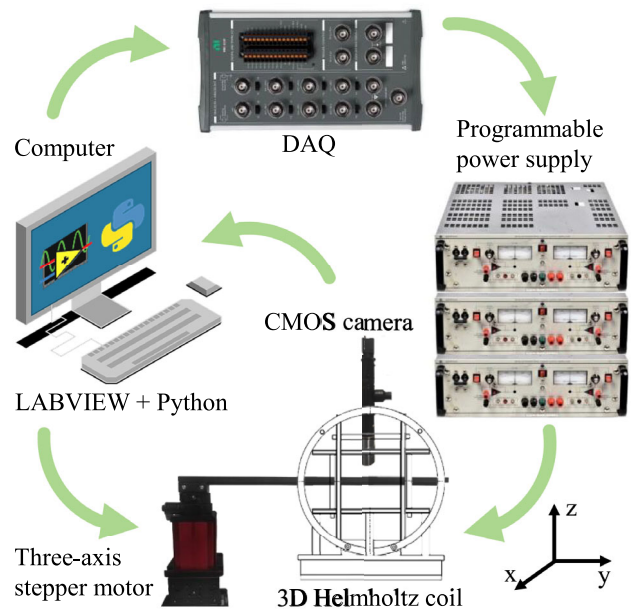


Figure 8. Components of the magnetic actuation system.

1) scanning the micro maze for imaging stitching and 2) aligning the FOV with the position of the microrobot during a control session.

A test chamber consisting of a $7 \times 7 \text{ mm}$ micromaze and PDMS walls was constructed. The maze was lithographically etched using SU-8 through ultraviolet light. The chamber was filled with 1.5% PVP solution. A drop of Fe_3O_4 nanoparticle suspension ($20 \mu\text{L}$, 1 mg mL^{-1}) was added to the 1.5% PVP solution inside the chamber via pipetting. The diameter of the nanoparticle was 800 nm , and the frequency of the visual feedback control was 20 Hz . Using the elliptical magnetic field with a frequency of 16 Hz and a strength of 8 mT , the dispersed magnetic nanoparticles form an elliptical swarm.

3.2. Validation of Image Stitching

The alignment of contours between the low-magnification image and the stitched image was evaluated, as shown in Figure 9. The low-magnification image, captured using a $1 \times$ objective lens, provides an overview of the map Figure 9(a) but suffers from low resolution, limiting the ability to observe the motion characteristics of the microrobot swarm (Figure 9(e)). Each pixel in the $1 \times$ magnification image represents information from three pixels in the $3 \times$ magnification image, resulting in insufficient detail.

In contrast, the stitched image is constructed from nine images captured using a $3 \times$ objective lens (Figure 9(b)). To compare the two, both images were resized to identical dimensions, and their contours were extracted after applying Gaussian blurring, binarization, and morphological opening. As shown in Figure 9(c), the contours exhibit a high degree of alignment. While some areas in the low-magnification image may appear slightly larger due to reflections, the discrepancy remains within

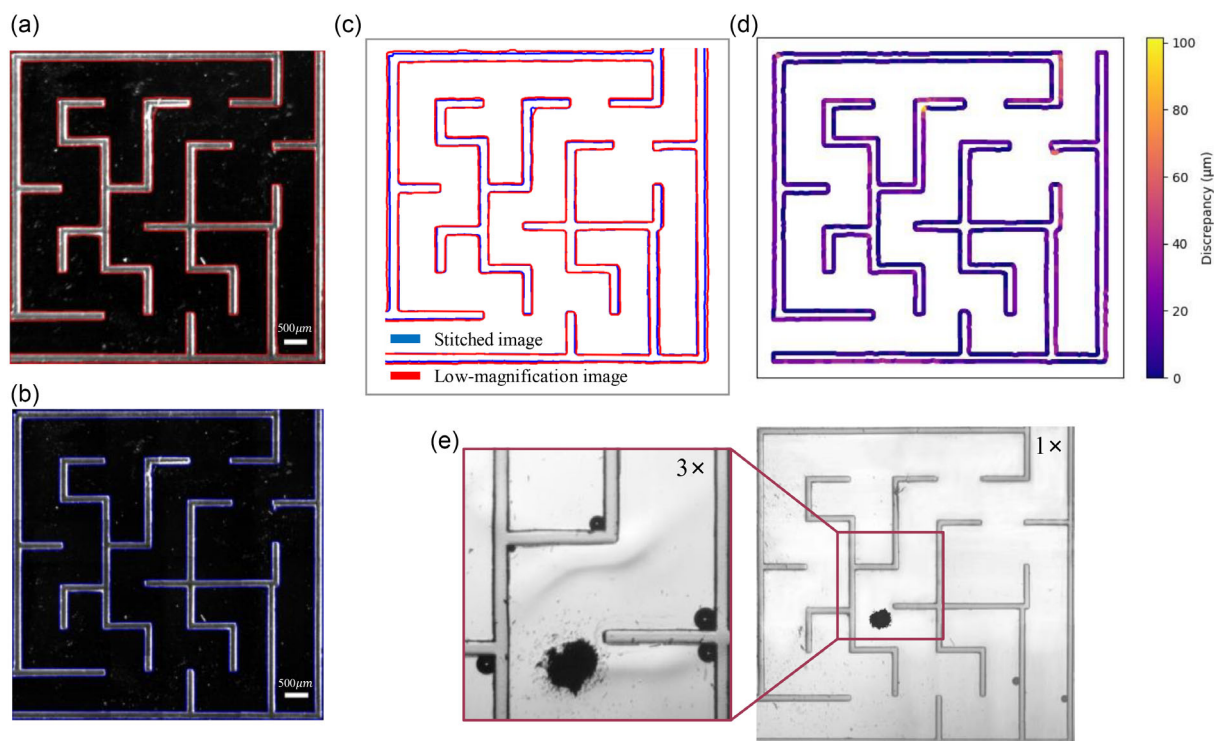


Figure 9. Comparison between the contours of the low-magnification image and the stitched image. a) Low-magnification image acquired with a $1\times$ lens. b) Stitched image formed by merging nine images captured with a $3\times$ lens. The scale bars represent $500\text{ }\mu\text{m}$. c) Contour alignment diagram. The blue contour corresponds to the stitched image, and the red contour corresponds to the low-magnification image. d) Contour alignment error plot. Color legend represents error. e) Comparison of swarms under $1\times$ and $3\times$ magnification.

$100\text{ }\mu\text{m}$, as illustrated in Figure 9(d). The average contour error was measured at $14.42\text{ }\mu\text{m}$, with a standard deviation of $12.26\text{ }\mu\text{m}$. These findings confirm that image stitching enables the creation of large-scale, high-precision maps suitable for swarm navigation. Given that the channel width was $1000\text{ }\mu\text{m}$ and the configuration space was accounted for, an error margin of 1% was negligible and does not significantly impact navigation accuracy.

3.3. Adaptive Deformation Through Channel

To validate the algorithm's capability to enable adaptive swarm deformation in a large workspace, the swarm was autonomously guided through a narrow-channel measuring $600\text{ }\mu\text{m}$ in width. The experimental results are presented in **Figure 10**.

Initially, the swarm formed a vortex shape in FOV 1. Using real-time threshold segmentation and contour extraction, the feedback images identified the channel ahead as narrower than the swarm. The swarm followed the planned path to the channel entrance, halting at 9 s, where it began deforming into an elliptical shape. During this process, the aspect ratio of the swarm increased progressively until the control system determined that the swarm's short axis was sufficiently narrow to pass through the channel. Once the swarm reached the required width, it resumed moving along the planned path, transitioning to FOV 2.

After traversing the narrow channel and entering FOV 3, the control system detected an increase in channel width. The elliptical swarm was then controlled to gradually revert to its vortex shape. Because the remaining section of the channel posed no collision risk, the swarm continued along the planned path without stopping for deformation.

The swarm achieved an average path-following error of $9.66\text{ }\mu\text{m}$. Despite an initial offset caused by control delays, this was quickly corrected by the visual feedback control system, as shown in Figure 10(b). The transition from vortex to elliptical shape occurred between 9 and 97 s, and the return to vortex shape occurred between 228 and 244 s. After completing the deformation at 97 s, a slight reduction in swarm width was observed, as depicted in Figure 10(c). This occurred due to inherent delays in swarm deformation, causing the width to stabilize only after the control input ceased.

3.4. Path Planning Through Maze

To validate the motion control, path planning, and FOV planning algorithms, the swarm was autonomously navigated through the maze to reach a designated endpoint. Experimental results of this navigation are presented in **Figure 11**. The global path length was $\approx 10\text{ mm}$, requiring the mobile platform to move three times and complete navigation within four FOVs over 267 s.

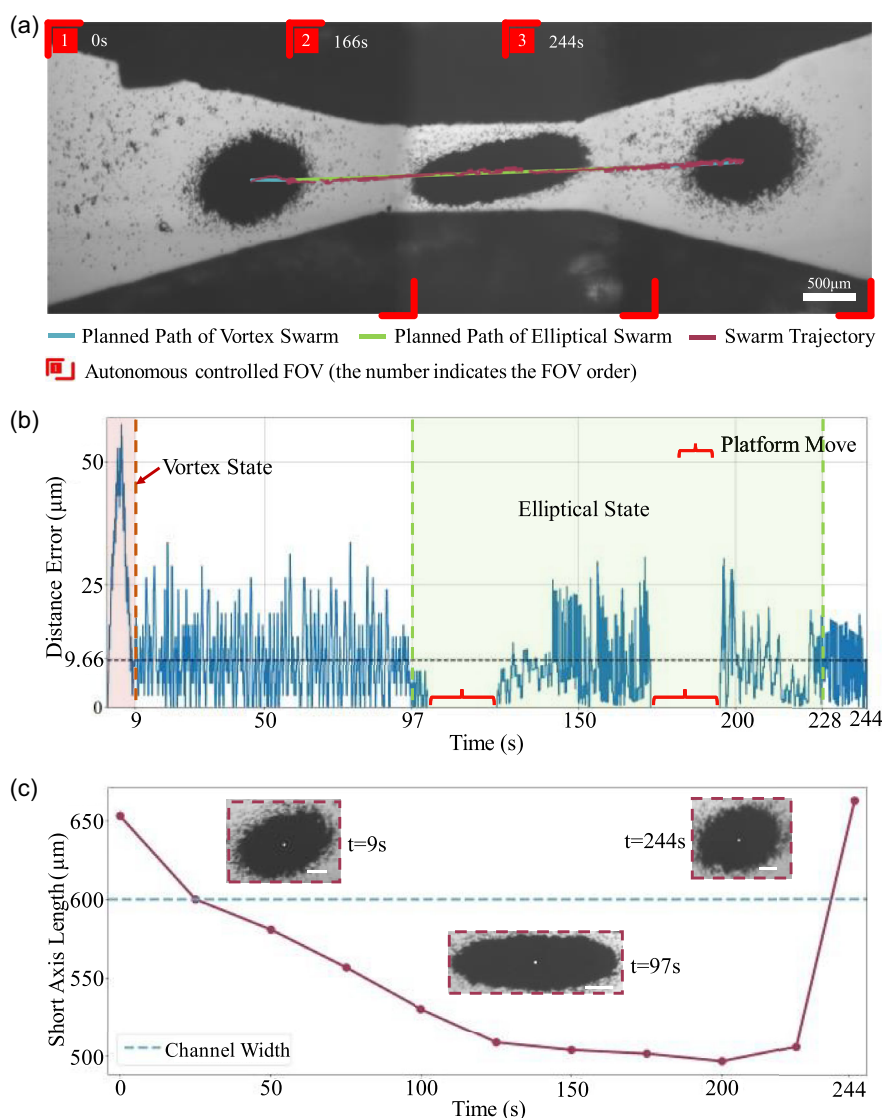


Figure 10. Adaptive motion of a microrobot swarm in a large workspace. a) Adaptive motion control of a microrobot swarm through a narrow channel across a large workspace. b) Distance error between the position of the swarm and the planned path. c) The plot of the swarm's short-axis length over time. The scale bars represent 500 μm.

A comparison of the global and local paths shows that the A* algorithm successfully identifies the nearest passage in the maze. In FOV 4, as shown in Figure 11a, the local path generated using OI-RRT* was noticeably smoother than the global path planned by the A* algorithm.

During the experiment, control alternated between swarm motion control and platform displacement. From 0 to 39 s, the swarm moved along the planned path in FOV 1. From 39 to 43 s, the platform shifted 853 μm to FOV 2. The swarm then navigated in FOV 2 from 43 to 136 s, followed by a 1951 μm platform shift to FOV 3 from 136 to 148 s. The swarm continued moving in FOV 3 from 148 to 198 s, and the platform finally moved 2682 μm to FOV 4 from 198 to 209 s. The swarm reached the endpoint in FOV 4 at 267 s. Throughout, the swarm maintained a nearly constant velocity of $37.54 \mu\text{m s}^{-1}$, as shown in Figure 11b.

Trajectory comparisons in Figure 11c reveal that swarm motion errors were more pronounced during turns, with average errors of 19.35 μm in FOV 2 and 17.11 μm in FOV 3. Conversely, errors were smaller during straight-line motion, averaging 8.62 μm in FOV 1 and 12.55 μm in FOV 4.

Experiments comparing navigation with and without the FOV planning algorithm demonstrate its effectiveness in improving efficiency and reducing swarm loss, as shown in Figure 11d. At a pitch angle of 1°, the swarm with the FOV planning algorithm reached the endpoint in 267 s, while the swarm without the algorithm took 341 s, demonstrating a 21% reduction in completion time. In both cases, swarm loss was ≈9%. When the pitch angle was increased to 2°, the swarm without the FOV planning algorithm completed the navigation in 204 s, a 23% improvement over the 1° case with the algorithm. However, swarm loss increased to 38%. These results underscore the importance of

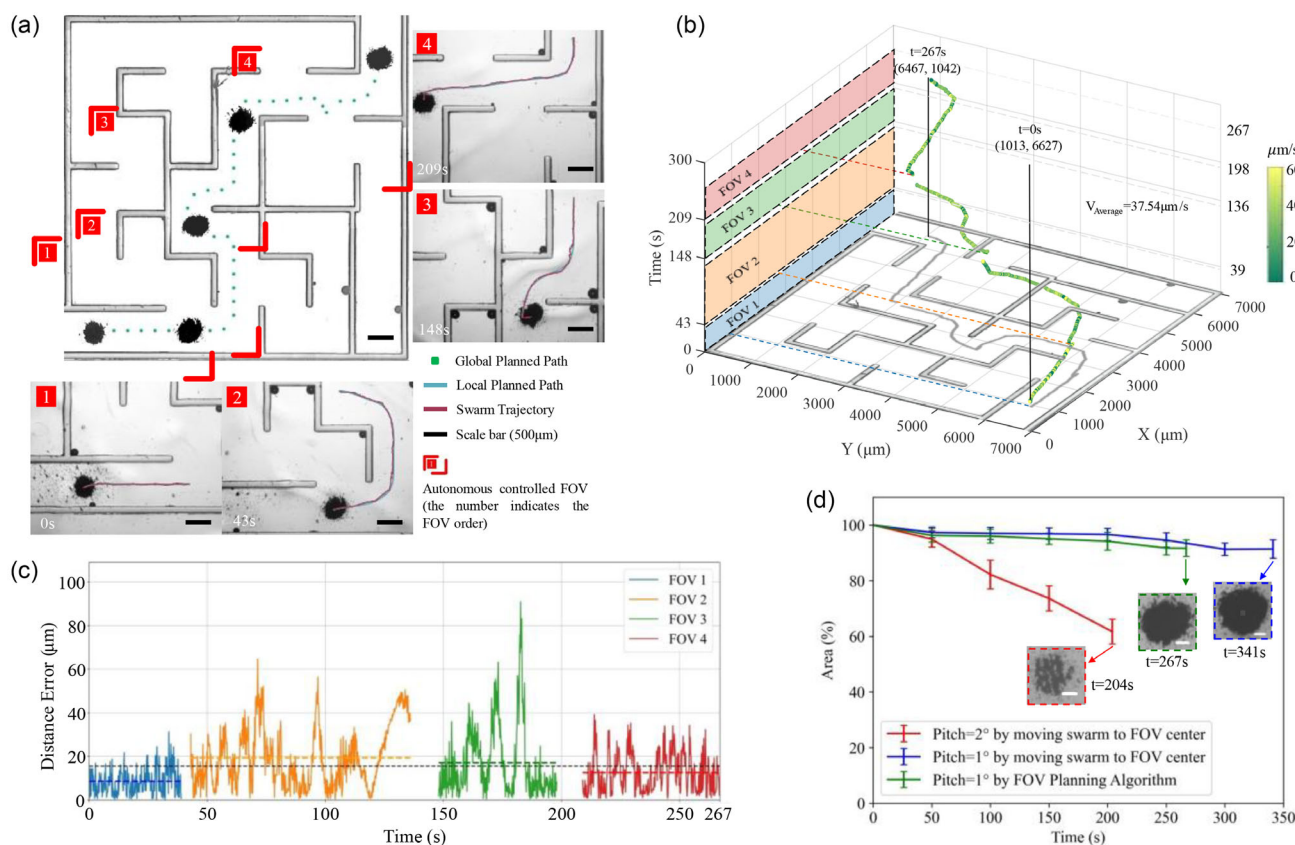


Figure 11. Navigation of a microrobot swarm in a large workspace with a micro maze. a) A* path global planning of the swarm in the global FOV, as well as OI-RRT* local path planning and motion trajectory in 4 local FOVs. b) The plots of trajectory over time in a large workspace with a micro maze. The times of 0 to 39 s, 43 to 136 s, 148 to 198 s, and 209 to 267 s correspond to FOV 1, FOV 2, FOV 3, and FOV 4 respectively. The times of 39 to 43 s, 136 to 148 s, and 198 to 209 s correspond with platform movement. c) The distance errors of the swarm. Different colors represent different FOVs. d) A comparative analysis of the area covered by the swarm over time, considering different pitch angles and various FOV movement algorithms within the large workspace. The scale bar represents 100 μm.

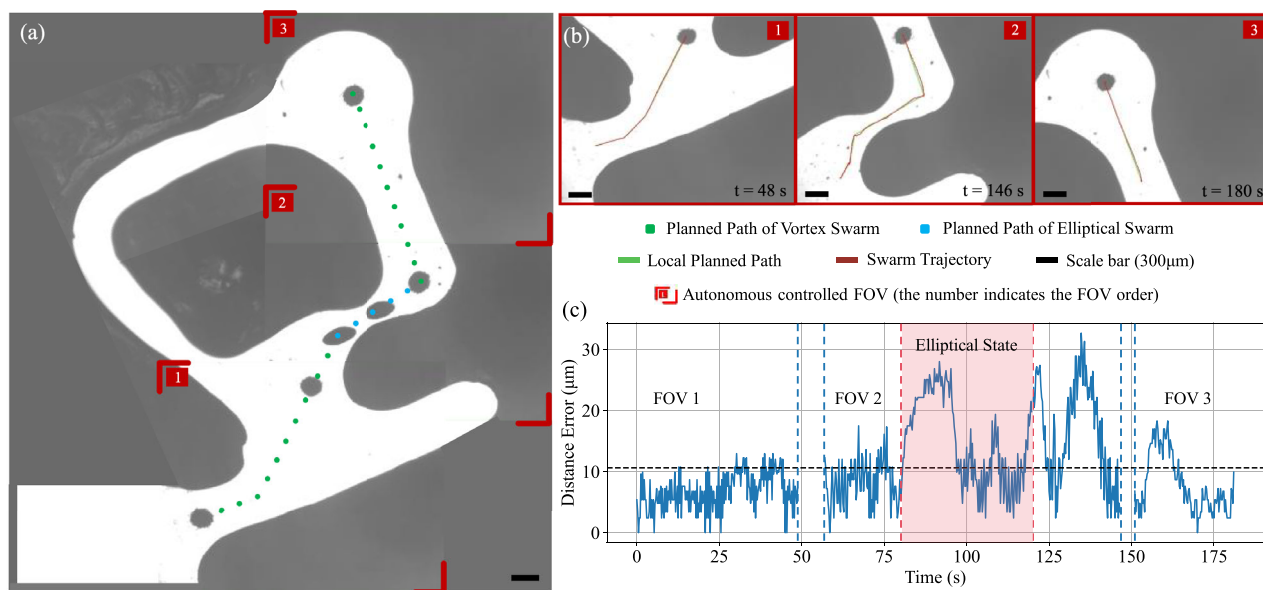


Figure 12. Deformable navigation through constrained environments. a) The global map of the channel with the FOVs and swarm trajectory. b) Detailed view of the three FOVs. c) Distance error during autonomous navigation.

the FOV planning algorithm in balancing motion efficiency and minimizing swarm loss.

3.5. Deformable Navigation through Constrained Environments

A maze-like channel with nonuniform widths was designed to assess the capability of the swarm to perform autonomous navigation and adaptive deformation across multiple FOVs in a large-scale, irregular environment (Figure 12(a)). The global path measures ≈ 6 mm in length and requires about 180 s to traverse, covering three FOVs from the starting point to the goal (Figure 12(b)). In the experiment, from 0 to 48 s, the swarm autonomously navigated the path determined by the OI-RRT* algorithm through the first FOV. From 58 to 146 s, the swarm traversed the second FOV, where it encountered a narrow passage. Once the swarm approached the entrance of the narrow path at 80 s, the elliptical magnetic field was adaptively adjusted under visual feedback control, enabling the microrobot to reduce its width and transition into an elliptical state to navigate through the confined space. After passing through at 120 s, the swarm reverted to the vortex formation and continued automatic navigation within the second FOV. From 151 to 180 s, the swarm moved through the third FOV and eventually reached the destination, as shown in Figure 12(b). Throughout the process, the swarm maintained an approximately constant speed of $33.34 \mu\text{m s}^{-1}$. The average path-following error was around $10 \mu\text{m}$ (Figure 12(c)), and the final loss rate after reaching the destination was only 2%, demonstrating good stability and adaptability.

4. Conclusion

In this article, we propose an effective and efficient swarm control scheme for the autonomous navigation of microrobot swarms within expansive workspaces that span multiple FOVs. Through the integration of phase correlation and template matching for image stitching, we successfully obtain extensive, high-resolution global images, effectively addressing the challenge of balancing image quality with FOV magnitude; thus enhancing the performance of path planning and improving the accuracy of visual feedback control for swarm trajectory tracking. Employing the A* algorithm for global path planning and OI-RRT* algorithm for local path planning ensures efficient and accurate navigation of the microrobot swarms. Furthermore, the FOV Planning algorithm optimizes the positioning of the FOV to reduce the number of movements and distance traveled by the displacement platform, thereby enhancing the efficiency and stability of swarm navigation. Integration of a 3-DOF displacement platform and feature point matching further refines navigation capabilities, enabling a seamless transition between local and global maps. Additionally, the implementation of a visual feedback control system enables real-time monitoring of channel width and swarm position, facilitating adaptive deformation and trajectory tracking of the microrobot swarms. Through experimental validation, we have demonstrated the efficacy and robustness of our automatic navigation control strategy in navigating large workspaces.

While the low-resolution image shown in Section 3 would've been sufficient for path planning, workspaces that are significantly larger will suffer from low resolution. It can be presupposed that certain paths and obstacles in significantly larger workspaces will be insufficiently resolved for path planning, leading to significant impacts on the quality of motion control. However, the limitation of the hardware used for this study lacked the necessary conditions to work with significantly larger workspaces. Nonetheless, the method presented herein can be applied to hardware that handles significantly larger workspaces and is unimpacted by the size of the workspace.

In future work, this study will be expanded to experiments closer to practical applications. The microchannels used for this work are ideal environments, lacking several realistic elements. The interior surfaces of the microchannel can be lined with biological tissues or membranes, and the fluid can be changed to biofluids in order to mimic biological environments. A* path planning was used here for its effectiveness, simplicity, and low computational cost; however, more advanced planning algorithms can be used to navigate these biomimicry environments. Normalized cross-correlation (NCC) can also be used in place of SSD in scenarios with more significant brightness variation (Figure S4). The control strategy can also be extended to controlling multiple swarms (Figure S3 and Video S7). Moreover, different medical imaging techniques, such as ultrasound doppler imaging, fluorescence imaging, and magnetic resonance imaging, can be utilized to test the potential in vivo autonomous navigation of swarms in large workspaces.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

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